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# Giridhar Pai

Cyber Security Student | SOC Analyst Aspirant | Blue Team | Digital Forensics

Dakshina Kannada, Karnataka, India | [www.giridharpai@gmail.com](mailto:www.giridharpai@gmail.com) | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

## Professional Summary

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Cyber security student with practical experience in blue teaming, digital forensics, cloud security, SIEM, web security, and secure application development. Built and contributed to internship, freelance, and hackathon projects involving authentication, encryption, logging, role-based access control, vulnerability scanning, and secure deployment. Strong interest in SOC operations, threat detection, incident analysis, and development of defense-oriented systems.

## Education

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### Yenepoya University

Jul 2023 – Jul 2026

BCA in Cyber Security, Ethical Hacking, and Digital Forensics with IBM

CGPA: 9.15

- Specialized in cybersecurity fundamentals, ethical hacking, digital forensics, and applied defense workflows.

### S N Moodbidri Polytechnic

Aug 2019 – May 2022

Diploma in Computer Science and Engineering

### SVS Vidyagiri English Medium School, Bantwal

2007 – 2016

Primary and Higher Schooling

## Technical Skills

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- **Security:** Threat Hunting, Threat Detection, SIEM, IDS/IPS, Digital Forensics, Risk Management, Security Audits
- **Cloud:** AWS Security, IAM, Secure Architecture, Secrets Manager, AWS Shield
- **Tools:** Splunk, Nmap, Nessus, Linux, SQL, SOAR concepts
- **Programming:** Python, Java, C++, C, JavaScript
- **Web:** HTML, CSS, Vanilla JavaScript, Supabase, Netlify
- **Platforms:** TryHackMe, GitHub, Overleaf, Kali Linux, Windows

## Certifications

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- AWS Certified Security – Specialty
- Google Cybersecurity Specialization
- Digital Forensics Essentials (EC-Council)
- Splunk Training – 10 Courses
- Deloitte Australia Cyber Job Simulation Internship
- TryHackMe – 13 Certifications

## Experience and Projects

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### Tejas Raksha – Automated Web Application Security Scanner

Mar 2026 – Apr 2026

- Engineered a high-performance, multi-tenancy web security scanner for the agriculture sector to identify 12+ OWASP Top 10 vulnerability categories, including SQL Injection, XSS, SSRF, and Command Injection.
- Architected a distributed microservices platform using FastAPI and Celery, with Redis for message brokering and PostgreSQL for persistent storage.
- Implemented a responsive frontend with React 18, Vite, and Tailwind CSS, featuring real-time scan progress updates through WebSockets and branded PDF report generation.
- Containerized all services with Docker Compose for one-click deployment and built a plugin-based vulnerability scanning workflow with false-positive reduction stages.
- Enabled enterprise-ready features such as scheduled scanning, trend analysis, CVE enrichment, and REST API integration with CI/CD pipelines including Jira, Slack, and email.
- **Key Technical Skills:** Python, FastAPI, SQLAlchemy, Alembic, Celery, Redis, Pydantic, React 18, Vite, Tailwind CSS, Zustand, Framer Motion, Axios, Docker, Docker Compose, Nginx, Linux, OWASP Top 10, Bandit, Semgrep, pytest, WebSocket monitoring.

### Kids Wonder Learn (KWL) – Freelance Project

2025 – 2026

- Built a browser-based educational platform for children aged 2–13+ with student, parent, teacher, and admin roles.
- Developed interactive learning modules including games, art tools, story content, login workflows, dashboards, progress tracking, and activity logging.
- Used HTML5, CSS3, Vanilla JavaScript, Supabase, and Netlify to design a modular and scalable front-end architecture.
- Planned backend integration with authentication, PostgreSQL schema design, role mapping, visitor logging, and secure access control.
- Added security-oriented design decisions such as hashed visitor tracking, password handling improvements, and audit-style logging.
- Documented deployment flow, data flow, configuration management, and migration path from localStorage to Supabase.

#### **Manufacturing Vault – Freelance Project**

Mar 2026 – Apr 2026

- Built a production-grade file encryption platform using AES-256-GCM and Argon2id key derivation with a zero-persistence architecture.
- Deployed on Azure using Purdue Model-aligned 4-tier VNet segmentation and Azure WAF v2 for perimeter defense.
- Achieved 100% pass rate on a 45-point security audit covering cryptography, network isolation, and operational logic.

#### **Kinetic Forge – Freelance Project**

Mar 2026 – Apr 2026

- Implemented mandatory TOTP-based MFA, Google OAuth 2.0 SSO, and role-based access control for supply chain users.
- Encrypted sensitive cargo telemetry, built a real-time shipment map, and created tamper-resistant audit logging.
- Defeated SQLi, XSS, CSRF, and brute-force attacks using logout rules and rate limiting.

#### **Vault Protocol – Freelance Project**

Mar 2026 – Apr 2026

- Designed a dual-authorization workflow for financial transactions with mandatory MFA re-authentication for auditors.
- Implemented idempotency keys using UUIDv7 and Redis TTL to prevent duplicate financial actions.
- Added encrypted PII storage, scrypt-based password hashing, and hash-chained audit logging.

#### **Retail Data Security – Freelance Project**

Mar 2026 – Apr 2026

- Architected a three-layer defense model combining Azure WAF v2, VNet isolation, and AES-256-GCM encryption.
- Implemented Argon2id-based key derivation, JWT authentication, Redis-cached keys, and encrypted blob storage.
- Validated the system using tamper tests and web attack resistance checks.

#### **Obsidian Pulse – Freelance Project**

Mar 2026 – Apr 2026

- Developed a healthcare security platform with dual-factor identity restoration, encrypted PHI handling, and real-time telemetry.
- Added brute-force resistance through quarantine logic, Redis-backed idempotent actions, IP blacklisting, and audit logging.
- Built for secure and latency-aware healthcare workflows.

#### **EdTech Security Framework – Freelance Project**

Mar 2026 – Apr 2026

- Implemented FIDO2/WebAuthn passkeys, TOTP fallback, and Google OAuth 2.0 SSO with defense-in-depth controls.
- Added restoration logic, login lockouts, IP blacklisting, encrypted secrets, and tamper-resistant audit verification.
- Designed to demonstrate secure passwordless authentication and audit integrity for educational systems.

#### **DefendX – Real-Time Threat Detection and Monitoring System**

2025

- Built as a hackathon project to demonstrate real-time threat monitoring concepts for enterprise security operations.
- Focused on alerting, monitoring, and defender-centric visibility for security teams.

#### **AI-Powered DeepFake Detection System**

2025

- Developed a project using heatmap analysis and metadata extraction to identify manipulated media.
- Presented as part of CIDECODE hackathon work and reported 99% accuracy in the project profile.

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## Future City – AI Urban Planning Application

2025

- Built for CODE SPRINT as an AI-powered smart city planning application.
- Demonstrated full-stack collaboration, problem solving, and product-style development.

## Password Strength Checker

2024

- Created a security utility for analyzing password robustness using rule-based validation.
- Demonstrated secure password awareness and user-friendly security feedback.

## Doctors Appointment Booking System

2023

- Built a web-based healthcare appointment booking system.
- Awarded Best Project Award at KAUSHALYAM 2022.

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## Hackathons and Competitions

- CIDECODE Hackathon – Consolation Prize Winner for DeepFake Detection System.
- IBM ICE DAY 2025 – 2nd Place in CTF Competition and 2nd Runner-Up in 24-hour Hackathon.
- CODE SPRINT – Finalist (Top 7 out of 65 teams) and Consolation Prize Winner.
- TryHackMe – Ranked in the Top 1% globally.

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## Achievements

- 23+ certifications and security-related learning milestones across cloud, SIEM, forensics, and cyber defense.
- Active contributor in cyber security communities and student initiatives.
- Completed practical learning through labs, CTFs, and hands-on tool usage.

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## Leadership / Community

- OWASP Yenepoya – Member, April 2024 – Present.
- Participated in security challenges and cybersecurity research/community activities.

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## Publications / Writing

- *The Aggressive Defender: A Technical Guide to Proactive Threat Hunting*
- *The Thin Digital Line: Why India's Defense Now Rests in Your Hands*
- OWASP Yenepoya Hackers Unveiled post and related cybersecurity reflections.

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## Additional Links

- **GitHub:** [github.com/Giri227](https://github.com/Giri227)
- **LinkedIn:** [linkedin.com/in/giridharpai227/](https://linkedin.com/in/giridharpai227/)
- **Portfolio:** [giri227.github.io/Giridharpai/](https://giri227.github.io/Giridharpai/)